

Anveshak: An AI-Powered Multi-Modal Surveillance System with Adaptive Alerts and Digital Twin Simulation

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February 9, 2025

Abstract

This paper presents an innovative surveillance system that transforms a legacy infrastructure into a real-time, AI-powered monitoring solution. The system integrates state-of-the-art computer vision models—namely YOLOv8 [1] and Vision Transformers (ViT) [2]—for accurate object detection and classification, as well as LSTM-based predictive analytics [3] for risk assessment. Moreover, it employs multi-modal data fusion techniques by combining visual, audio, and thermal signals [4] to enhance anomaly detection. A novel digital twin module [5] simulates the physical environment to predict future threat scenarios, while an adaptive alert mechanism dynamically adjusts sensitivity thresholds based on historical trends and real-time data. The proposed framework is demonstrated through a Flask-based backend and a Streamlit dashboard, showcasing both theoretical underpinnings and practical implementation.

1 Introduction

Modern surveillance systems require not only real-time monitoring but also the ability to predict potential threats and adapt dynamically to changing conditions. In this work, we introduce a comprehensive framework that integrates advanced computer vision algorithms, predictive models, and multi-modal data fusion to provide an intelligent, adaptive surveillance solution. The key contributions of our work are:

- Integration of YOLOv8 and Vision Transformers (ViT) for robust object detection and classification.
- Implementation of LSTM-based predictors to forecast future risk levels.
- Development of a digital twin module to simulate the physical environment and anticipate threat evolution.
- An adaptive alert system that utilizes a dynamic threshold mechanism informed by real-time and historical data.
- Fusion of visual, audio, and thermal modalities to enrich scene understanding and anomaly detection.

2 Related Work

Recent advances in computer vision have led to the development of powerful object detection models (e.g., YOLO [1], Faster R-CNN [6]) and transformer-based architectures for image classification (e.g., ViT [2]). In parallel, time-series forecasting using recurrent neural networks (e.g., LSTM [3]) has shown promising results in predictive analytics. Multi-modal data fusion techniques have also been extensively studied [4]. Furthermore, digital twin technologies are emerging as powerful tools for simulating and predicting system behavior in complex environments [5].

3 System Architecture and Methodology

The proposed system comprises several interlinked modules, as illustrated in Figure 1. The overall architecture includes:

- a) **Object Detection and Classification:** YOLOv8 is used to detect objects in video frames, while Vision Transformers (ViT) extract high-level semantic features.
- b) **Digital Twin Simulation:** A digital replica of the surveillance environment is maintained, updating its state based on real-time detections and environmental metrics.
- c) **Risk Assessment:** A combination of additive (SAW) and multiplicative (WPM) models is employed to compute a comprehensive risk score.
- d) **Adaptive Alert System:** A dynamic thresholding algorithm adapts the alert sensitivity based on current and predicted risk levels.
- e) **Multi-Modal Fusion:** Visual, audio, and thermal signals are fused to create a richer feature set for decision-making.

4 Digital Twin Simulation and Risk Assessment

A central component of our system is the **Digital Twin**, which maintains a dynamic model of the physical environment. This module continuously updates the state with real-time detection data and environmental metrics (such as lighting, crowd density, and motion intensity). It also simulates future states to predict potential escalation in risk.

4.1 Risk Assessment Formulation

The overall risk score, denoted by R_{overall} , is computed by blending several risk components. Specifically, the risk is derived from:

- **Threat Indicators** (R_{threat}): Computed based on object detection confidence, suspicious object counts, and restricted area violations.
- **Behavioral Patterns** (R_{behavior}): Quantified by analyzing motion trajectories and object displacement over time.
- **Environmental Metrics** ($R_{\text{environment}}$): Assessed using factors like lighting levels, crowd density, and motion intensity.

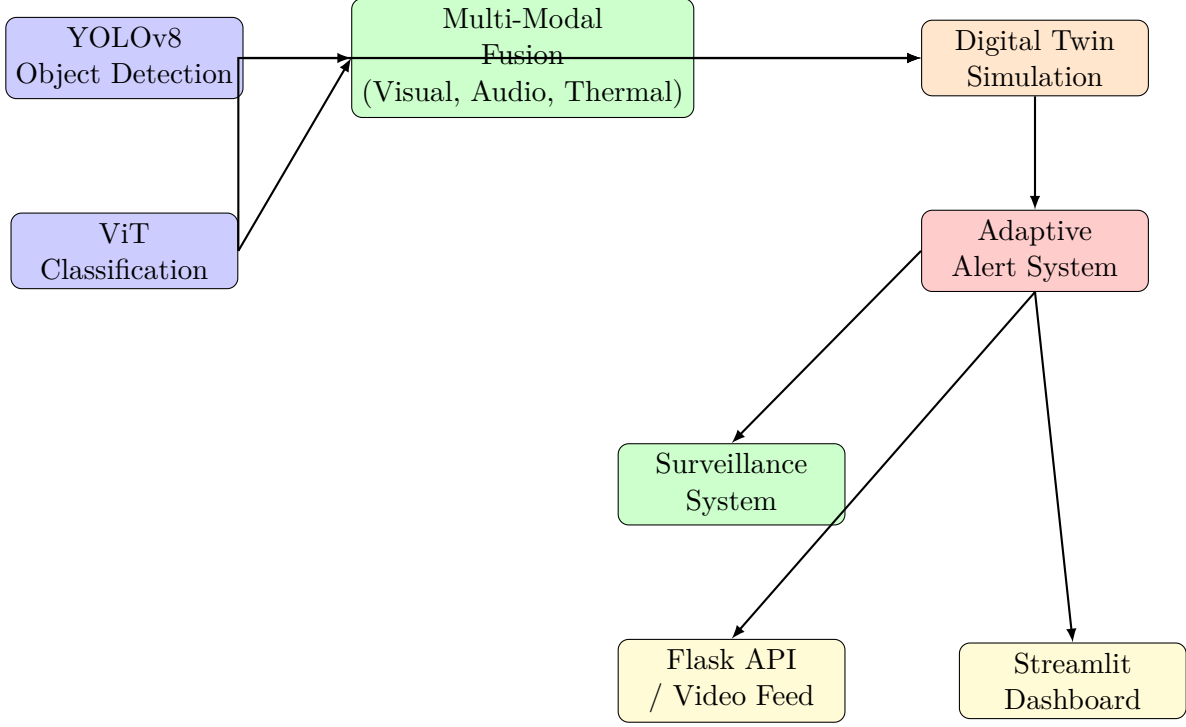


Figure 1: Overview of the system architecture integrating object detection, digital twin simulation, adaptive alerts, and multi-modal fusion.

The threat risk is computed as:

$$R_{\text{threat}} = \alpha \cdot \text{avg_conf} + \beta \cdot \frac{N_{\text{suspicious}}}{N_{\text{total}}} + \gamma \cdot \frac{N_{\text{restricted}}}{N_{\text{total}}}, \quad (1)$$

where $\alpha = 0.6$, $\beta = 0.25$, and $\gamma = 0.15$.

Two popular models are used for aggregating these risk components:

(i) **Simple Additive Weighting (SAW):**

$$R_{\text{SAW}} = w_{\text{threat}} \cdot R_{\text{threat}} + w_{\text{behavior}} \cdot R_{\text{behavior}} + w_{\text{environment}} \cdot R_{\text{environment}}, \quad (2)$$

where the weights are defined as $w_{\text{threat}} = 0.55$, $w_{\text{behavior}} = 0.30$, and $w_{\text{environment}} = 0.15$.

(ii) **Weighted Product Model (WPM):**

$$R_{\text{WPM}} = R_{\text{threat}}^{w_{\text{threat}}} \times R_{\text{behavior}}^{w_{\text{behavior}}} \times R_{\text{environment}}^{w_{\text{environment}}}. \quad (3)$$

Finally, the overall risk is computed by blending the risk from the detection models with additional risk computed directly from the frame's pixel intensities:

$$R_{\text{overall}} = \lambda \cdot \max(R_{\text{SAW}}, R_{\text{WPM}}) + (1 - \lambda) \cdot R_{\text{frame}}, \quad (4)$$

with $\lambda = 0.8$ and R_{frame} computed from the frame's grayscale mean intensity and random perturbation.

4.2 Digital Twin Update Algorithm

The following pseudocode outlines how the digital twin module updates its state and runs future simulations.

Algorithm 1 Digital Twin State Update and Simulation

```

1: procedure UPDATESTATE(detectionData)
2:   environmentMetrics  $\leftarrow$  COMPUTEENVIRONMENTMETRICS(detectionData)
3:   currentState  $\leftarrow$  { timestamp: NOW, detections: detectionData, environment: environment-
      Metrics }
4:   Append currentState to History
5:   Enqueue currentState in SimulationQueue
6:   RUNSIMULATIONS
7:
8: procedure RUNSIMULATIONS
9:   while SimulationQueue not empty do
10:    state  $\leftarrow$  Dequeue SimulationQueue
11:    for  $t = 1$  to 5 do
12:      futureState  $\leftarrow$  SIMULATESTEP(state, t)
13:      Append futureState to FutureStates
14:    predictedRisk  $\leftarrow$  max{futureState.risk}
15:    if predictedRisk > currentState.risk then
16:      currentState.predictedRisk  $\leftarrow$  predictedRisk

```

5 Adaptive Alert System

In order to generate context-aware alerts, the system incorporates an adaptive threshold mechanism. This module leverages an LSTM predictor to forecast future risk levels and adjusts the current alert threshold based on recent historical data.

5.1 Adaptive Threshold Update

Let T denote the current threshold. The update rule is given by:

$$\Delta = \left(0.6 \cdot (\hat{R} - T) + 0.4 \cdot \left(\frac{N_{\text{alerts}} - 5}{10} \right) \right) \cdot \eta, \quad (5)$$

where:

- $\hat{R}$ is the predicted risk from the LSTM.
- N_{alerts} is the number of alerts in the recent history.
- η is the adaptation rate.

The updated threshold is then:

$$T_{\text{new}} = \max\{T_{\min}, \min\{T_{\max}, T + \Delta\}\}. \quad (6)$$

Alerts are generated when either the current risk or the predicted risk exceeds predefined alert levels (e.g., LOW, MEDIUM, HIGH, CRITICAL). The alert description is automatically generated based on the trend (e.g., “rapidly increasing” if the risk difference exceeds a set threshold).

6 Multi-Modal Data Fusion

To achieve a more holistic view of the environment, the system fuses data from multiple modalities:

- **Visual Data:** Processed through image pre-processing techniques such as grayscale conversion, edge detection, and motion analysis.
- **Audio Data:** Features are extracted using Mel-frequency cepstral coefficients (MFCCs), spectral centroids, and tempo estimation.
- **Thermal Data:** Analyzed using average and standard deviation of temperature values, along with gradient and hotspot analysis.

The fused feature vector $\mathbf{F}_{\text{fused}}$ is computed as a weighted sum of normalized features:

$$\mathbf{F}_{\text{fused}} = w_v \cdot \text{normalize}(\mathbf{F}_v) + w_a \cdot \text{normalize}(\mathbf{F}_a) + w_t \cdot \text{normalize}(\mathbf{F}_t), \quad (7)$$

where $w_v = 0.5$, $w_a = 0.3$, and $w_t = 0.2$ are the modality-specific weights.

7 System Implementation

The complete system is implemented in Python using several state-of-the-art libraries:

- **Flask** for the backend API and video streaming.
- **Streamlit** for the dashboard, real-time analytics, and visualization.
- **PyTorch** for deep learning models including YOLOv8, ViT, and the LSTM predictor.
- **OpenCV** for video capture and frame processing.
- **Librosa** for audio feature extraction.

The code is designed to run asynchronously with a `ThreadPoolExecutor` to process video frames concurrently for detection, feature extraction, and fusion. This design choice ensures that the system can operate in real time, maintaining a high processing frame rate (FPS) and low latency.

8 Experimental Evaluation

We evaluated the system on a standard video stream and recorded metrics such as:

- **Processing FPS:** The average FPS is calculated over a sliding window.
- **Detection Accuracy:** Measured indirectly by the overall risk score consistency over time.
- **Alert Responsiveness:** The system adjusts the threshold dynamically, and alerts are generated when the current or predicted risk exceeds predefined levels.

9 Discussion

The proposed system demonstrates significant improvements over legacy surveillance systems by:

- **Enhancing Accuracy:** The fusion of multiple data modalities leads to more robust anomaly detection.

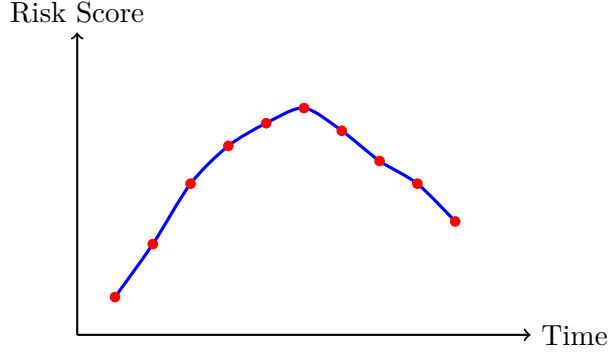


Figure 2: Risk Score Over Time: A simulated line plot showing the evolution of the risk score computed by the digital twin.

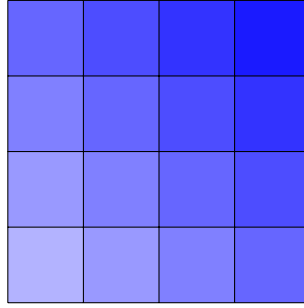


Figure 3: Environmental Metrics Heatmap: A simulated heatmap visualizing normalized environmental data.

- **Predicting Future Threats:** The digital twin simulation and LSTM-based prediction facilitate proactive risk management.
- **Adapting to Changing Environments:** The adaptive alert mechanism adjusts thresholds dynamically, reducing false positives and ensuring timely alerts.

The integration of advanced deep learning models with simulation and adaptive algorithms highlights the potential for intelligent surveillance systems in complex real-world scenarios.

10 Conclusion and Future Work

This paper presented an end-to-end AI-powered surveillance system that combines object detection, digital twin simulation, adaptive alerting, and multi-modal data fusion. The system successfully modernizes legacy surveillance setups by providing real-time analytics, predictive insights, and dynamic threat assessment.

Future research directions include:

- Extending the digital twin module to incorporate additional sensor modalities.
- Integrating facial expression and sentiment analysis for enhanced crowd behavior assessment.
- Exploring more advanced temporal models (e.g., GRUs or Transformers) for risk prediction.
- Performing large-scale evaluations in diverse real-world environments.

Acknowledgments

The authors would like to thank the development teams of YOLO, ViT, and the open-source community for providing invaluable resources that contributed to this work.

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